Stationary points

1 State the type of point that matches the following descriptions:

a A point where the curve changes from decreasing to increasing.

b A point where the curve changes from increasing to decreasing.

c A point where the tangent is horizontal and the concavity changes about the point.

2 For each of the following functions:

i Find the derivative.

ii Find the coordinates of any stationary points.

iii Classify each stationary point.

a $y = -6x^2 + 84x - 29$

b $y = x^3 - 21x^2 + 144x - 19$

c $f(x) = (x+3)^2(x+6)$

d $f(x) = (x+5)^3 + 4$

e $f(x) = -\frac{x^3}{3} + \frac{13x^2}{2} - 30x + 10$

f $f(x) = (4x+5)(x+1)$

g $f(x) = 134 - 300x + 240x^2 - 64x^3$

h $f(x) = (x^2 - 9)^2 + 4$

3 Consider the parabola with equation $y = 5 + x - x^2$.

a Find the coordinates of the vertex of the parabola.

b State the gradient of the tangent to the parabola at the vertex.

c What type of stationary point is at the vertex of this parabola?

4 Consider the function $f(x) = x^2 + 4x + 9$.

a Find an equation for the gradient function $f'(x)$.

b State the interval in which the function is increasing.

c State the interval in which the function is decreasing.

d Find the coordinates of the stationary point.

e Classify the stationary point.

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5 Consider the function $f(x) = 4x^3 + 5x^2 - 4$.

a Find an equation for the gradient function $f'(x)$.

b Find the coordinates of the stationary points.

c Complete the table of values:

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$		0		0	

d Hence determine the:

i Local minimum

ii Local maximum

e Is -4 the absolute minimum value of the function? Explain your answer.

6 Consider the function $f(x) = 3x^2 - 54x + 241$.

a Find $f'(x)$.

b Find the x-coordinate of the stationary point.

c Classify the stationary point.

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